A1
PICO 2 SC1632 SOCKETED
DOCK_TX
DOCK_RX
MCU_MOTOR_A_N
MCU_DS_B_N
MCU_DS_A_N
MCU_MOTOR_B_N
MCU_DIR_N
MCU_STEP_N
MCU_WDATA_N
MCU_WGATE_N
MCU_HEAD_N
MCU_DENSEL
ASSERT_INDEX
ASSERT_TRK0
ASSERT_WP
ASSERT_RDATA
ASSERT_CHANGE
SPI_CS_N
SPI_SCK
SPI_MOSI
SPI_MISO
MEDIA_ARM
INTERLOCK_OK
MEDIA_ERROR_N
LED_DATA
DOCK_BUSY
V3V3
VSYS
J1
34-PIN PC FLOPPY
FDD_DENSEL
FDD_INDEX_N
FDD_MOTOR_A_N
FDD_DS_B_N
FDD_DS_A_N
FDD_MOTOR_B_N
FDD_DIR_N
FDD_STEP_N
FDD_WDATA_N
FDD_WGATE_N
FDD_TRK0_N
FDD_WP_N
FDD_RDATA_N
FDD_HEAD_N
FDD_CHANGE_N
U1
SN74LVC245AN / 5V→3V3, INPUT
U2
SN74LVC245AN / 5V→3V3, INPUT
J2
FLOPPY POWER 5V GND GND 12V
Q4
TN0702 / FDD INDEX
Q5
TN0702 / FDD TRK0
Q6
TN0702 / FDD WP
Q7
TN0702 / FDD RDATA
Q8
TN0702 / FDD CHANGE
RN1
8x10k BUSSED / COMMON PINS
RN2
8x10k BUSSED / COMMON PINS
RN3
8x1k BUSSED / COMMON GND
F1
2A FAST / LOW DROP
D1
1N5819 / USB BACKFEED BLOCK
C1
47uF 16V
C2
100nF 50V
C3
100nF 50V
C5
100nF 50V
J3
ISOLATED CONTACTS / 1 TO 7
J4
DOCK LINK 3V3 UART
R1
10k SPI CS DEFAULT
R2
10k MEDIA ERROR PULLUP
J5
3V3 OUT / INTERLOCK RETURN
U4
SN74LVC245AN / SWITCHED MEDIA ISOLATION
U6
LP2951ACN / SWITCHED 3V1 100mA
R4
10k 1%
R5
10k 1%
R6
68R FAILSAFE BLEED
R7
680R DEFAULT OFF
R8
2R2 DISCHARGE
R9
10k MSO DEFAULT
R10
1 OHM CAP ESR
R11
2k2 ARM GATE
R12
2k2 INTERLOCK GATE
Q1
IRLZ44N / HARD DISCHARGE
Q2
TN0702 / ARM PERMISSION
Q3
TN0702 / INTERLOCK PERMISSION
C6
22uF 16V / OUTPUT
C7
100nF 50V
C9
1uF 50V X7R
Q9
SUP53P06 / MEDIA INPUT CUT
Q10
TN0702 / OE ARM
Q11
TN0702 / OE INTERLOCK
R13
10k OE DEFAULT OFF

R1: Standard IBM-PC 34-pin FDD rear interface; cable twist handled by firmware selection.
5V inputs use 5.5V-tolerant LVC245 at 3.3V; Outputs are TN0702 open drain: ASSERT=1 pulls cable LOW.
J3 is the isolated seven-wire contact output. J5 series seat-NO/eject-NC loop MUST open before disk motion.
Pico is a factory-assembled SMD module plugged into TH sockets. Carrier assembly is through-hole.
Legacy floppy encoding/track streaming and native dock protocol require new firmware; no compatibility claim.

PROTOTYPE / NOT RELEASED / electrical and mechanical bench tests required

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Title: RODDY CYBER DRIVE / CORE CONTROLLER

Size: A1	Date:	Rev: R1.1 hardware finish
KiCad E.D.A. 10.0.6		Id: 1/1